

Performance Evaluation and Benchmarks

This document reports the empirical benchmarking results comparing key-value storage engines under high concurrency workloads.

Configuration Matrix

The experimental evaluation parameters are summarized in the table below:

Component	Default Value	Description
Connection Pool	64 connections	Maximum concurrent database connections per node
Write Buffer	128 MB	In-memory write ahead log buffer capacity
Heartbeat Interval	500 ms	Consensus leader ping interval for failover detection
Compaction Threads	4 threads	Background threads dedicated to LSM-tree SSTable compaction

Architecture Overview

The following diagram illustrates the cluster topology and data synchronization path between active replicas:

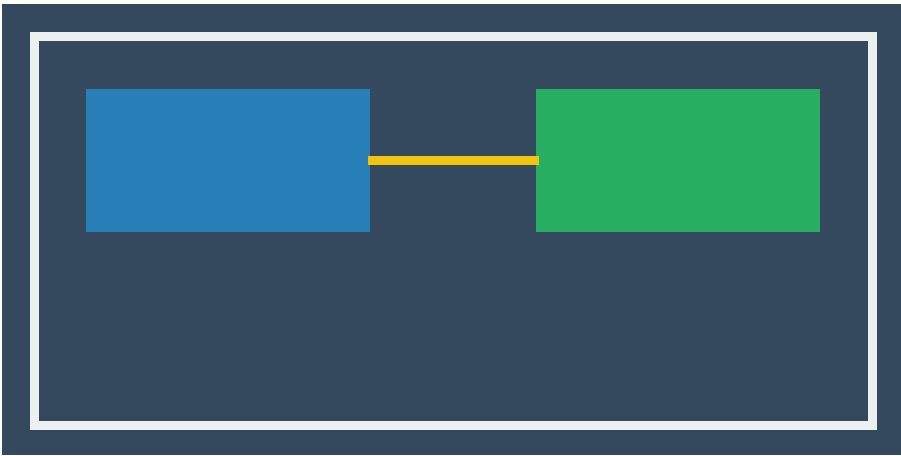


Figure 1: Cluster replication and failover architecture diagram

All replica nodes stream write operations asynchronously while retaining quorum verification for consensus.